

Do On-Chain Signals Improve Bitcoin Market-Regime Classification?

A Supervised Learning Approach with Honest Evaluation of Data Limitations

CPSC 381/581: Machine Learning | Yale University | Instructor: Alex Wong

1. Name

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2. Problem Statement

Bitcoin markets cycle through three qualitative regimes recognized by both practitioners and academic researchers: **Accumulation** (price consolidation, informed buyers absorbing supply), **Expansion** (sustained directional trend), and **Distribution** (price plateau, early holders reducing exposure). Correctly identifying the current regime informs position sizing, risk limits, and entry/exit timing.

Standard approaches rely on price and volume alone. A distinct class of signals — **on-chain data** — records the actual transfer behavior of Bitcoin holders directly on the public blockchain. The central research question is: *do on-chain features carry regime-predictive information beyond what price-derived features already encode?* This is formulated as a supervised **multi-class classification** problem: given a feature vector constructed at time t , predict the regime label $y \in \{Accumulation, Expansion, Distribution\}$ characterizing the market over the subsequent 30-day window.

Why this problem is non-trivial:

- Financial time series violate i.i.d. assumptions: observations are serially correlated, distributions shift across cycles, and class frequencies are highly imbalanced.
- The mapping from on-chain activity to regime is indirect — it is not obvious a priori that any ML model can extract a reliable signal, making null results equally informative.
- The problem requires confronting label construction as a modeling choice, not a given — a subtlety absent from standard benchmark datasets.

Acknowledged scope limitations:

■ *On-chain data captures only spot-market holder behavior. Short-term Bitcoin price volatility is largely driven by the derivatives market (perpetual futures, options), which leaves no on-chain trace. This project therefore targets medium-term regime identification (30-day horizon), not short-term price prediction.*

■ *If a model consistently predicts regime transitions, rational market participants will trade against its signals, eroding any predictive edge over time. The academic contribution here is the empirical test of signal existence — not a claim of persistent, deployable alpha.*

3. General Approach

3.1 Label Construction — the hardest part

Ground-truth regime labels do not exist and must be constructed. This is a deliberate modeling choice with significant downstream consequences, handled in two steps:

- **Rule-based initialization:** Use the Market-Value-to-Realized-Value (MVRV) ratio as a financially motivated prior. $MVRV < 1.0 \rightarrow$ Accumulation (holders are on average underwater, historically associated with cycle troughs); $MVRV > 3.5 \rightarrow$ Distribution (large unrealized profits, historically associated with cycle tops). Intermediate periods are labeled Expansion. Thresholds are calibrated against documented Bitcoin cycle peaks (Dec 2013, Dec 2017, Apr/Nov 2021) and troughs (Jan 2015, Dec 2018, Nov 2022).
- **Sensitivity analysis:** Labels will be regenerated under three threshold variants (conservative / moderate / aggressive). Results robust to all three are treated as credible; results that flip sign across variants are flagged as artifact-dependent.

■ *Critical design constraint: MVRV is used both to construct labels and as a feature. To prevent circular reasoning, an ablation will remove MVRV from the feature set and re-evaluate, verifying that other on-chain signals retain predictive power independently.*

3.2 Model Progression

Four model classes are implemented in sequence, each mapping directly to a course topic. The progression from simple to complex enables ablation: if a linear model achieves comparable performance to a neural network, the added complexity is unjustified.

Stage	Model	Course Connection	Role in Study
1	Ridge Regression (regression sub-task: 30-day fwd return)	Weeks 3, 9 — linear regression, regularization	Establishes price-predictability baseline; tests whether any return signal exists
2	Linear SVM (regime classification)	Week 11 — SVM, kernel methods	Linear regime-boundary baseline; L2 regularization via C parameter
3	Random Forest + XGBoost (classification)	Week 7 — nonlinear mapping; ensemble methods	Primary nonlinear model; SHAP values provide feature attribution across folds
4	1-D Temporal CNN (PyTorch, classification)	Week 13 — neural networks, SGD, momentum	Captures local temporal patterns across 30-day feature sequences

3.3 Walk-Forward Expanding-Window Evaluation

Standard k-fold cross-validation leaks future information into training folds for time series. An expanding-window walk-forward scheme is used instead: the model trains on all data up to halving cycle k , then is evaluated on cycle $k+1$. With four documented cycles (2013–15, 2016–18, 2019–21, 2022–26), this yields three non-overlapping test folds, producing a reliable estimate of out-of-sample generalization.

■ *Early folds have substantially fewer training observations (~300–500 days) than later folds (~2,000 days). Model variance is reported per fold — not just averaged — to expose this instability honestly rather than masking it in a single aggregate metric.*

3.4 Anticipated Challenges

- **Class imbalance:** Distribution phases occupy ~10–15% of trading days. Class-weighted cross-entropy loss and SMOTE oversampling will be compared; macro-F1 is the primary metric because it penalizes models that ignore minority classes.
- **Non-stationarity:** Features such as hash rate grow monotonically. Rolling z-score normalization within each training window is applied to mitigate distributional shift.
- **Lookahead bias:** All features use only past data; rolling computations are lagged appropriately. Code includes an explicit assertion verifying no future data enters any training sample.

4. Data

The dataset spans **January 2015 – March 2026** (~4,100 daily observations), covering three complete Bitcoin halving cycles. Two freely accessible sources are merged on the UTC date index:

Source	Variables	Economic Rationale	Access
CoinMetrics Community API	MVRV Ratio, Realized Cap, NVT Ratio, Coin Days Destroyed, Active Addresses, Hash Rate, Puell Multiple, SOPR	Direct behavioral signal: records what holders are actually doing with coins, not just what price is doing	Free REST API; bulk CSV at coinmetrics.io/community-data
Yahoo Finance (yfinance)	Daily OHLCV, 30-day realized volatility, 14-day RSI, MACD signal, BTC dominance	Price-only baseline features; enables ablation isolating on-chain marginal value	yfinance Python library; no API key required

Key ablation design: Models are trained and evaluated on two feature sets separately — (A) price-only, and (B) price + on-chain. The F1 delta between A and B is the direct empirical answer to the core research question. A null or negative delta is reported as an honest and important finding, not a failure.

■ *CoinMetrics on-chain data is available from ~2010, but several metrics (e.g., SOPR) have reliable coverage only post-2018. Features with short history are excluded from early folds to avoid imputing missing values that could introduce bias.*

5. Related Work

This project sits at the intersection of three verified lines of research:

- **Bitcoin market regime structure.** Ardia, Bluteau, and Rüede (2019) apply Markov-Switching GARCH models to Bitcoin daily log-returns and find strong statistical evidence of at least two distinct volatility regimes, showing that single-regime GARCH models systematically underperform in Value-at-Risk forecasting. Their finding motivates this project: if Bitcoin returns are structurally regime-dependent, a supervised classifier trained on leading indicators may identify regime transitions before they are reflected in realized volatility. Critically, Ardia et al. use an unsupervised generative model (HMM-GARCH); this project instead adopts a discriminative supervised approach with financially interpretable on-chain features as inputs.

- **On-chain data as a predictive signal.** A 2025 study published in *Expert Systems with Applications* (ScienceDirect) trains SVM, Random Forest, Gradient Boosting, and deep learning models using 92 on-chain metrics alongside 138 technical indicators for Bitcoin price direction and magnitude prediction, finding that on-chain data significantly improves classification performance over price-only baselines — and that the Boruta-selected SVM produces the best risk-adjusted backtest returns. This study is the closest existing work to the present project; the present project differs by (a) using a three-class regime taxonomy rather than binary up/down direction, (b) applying walk-forward expanding-window evaluation rather than a single train/test split, and (c) explicitly ablating the marginal value of on-chain features via a controlled price-only comparison.
- **ML classifier benchmarks for financial classification.** Ballings, Van den Poel, Hespeels, and Gryp (2015) benchmark ensemble methods (Random Forest, AdaBoost, Kernel Factory) against single classifiers (Neural Networks, Logistic Regression, SVM, KNN) for stock price direction prediction across 5,767 European companies, finding that Random Forest is the top-performing algorithm. This project replicates their comparative methodology — testing a hierarchy of models from linear to ensemble to neural — but applies it to a time-series multi-class classification task with walk-forward evaluation, rather than cross-sectional AUC.

Taken together, prior work establishes that (1) Bitcoin returns exhibit documented regime structure, (2) on-chain features improve classification beyond price-only baselines in at least one large-scale study, and (3) ensemble methods tend to outperform linear classifiers on financial data. This project tests whether all three findings jointly transfer to medium-horizon regime identification.

6. Measure of Success

Primary metric — Classification (regime):

- Macro-averaged F1 score across three classes. Macro-averaging treats each class equally regardless of frequency, directly penalizing models that ignore the minority Distribution class.
- **True baselines:** A stratified-random classifier on empirical class frequencies (~55% Expansion, ~30% Accumulation, ~15% Distribution) yields macro-F1 ≈ 0.31 in expectation. A majority-class classifier yields macro-F1 ≈ 0.28 . All models are compared against these, not against an arbitrary round number.
- Success threshold: Linear SVM macro-F1 ≥ 0.45 (substantially above chance); best nonlinear/neural model macro-F1 ≥ 0.58 . Both targets are conservative given the genuine difficulty of the problem.

Secondary metric — On-chain marginal value:

- The F1 delta between model (B) [price + on-chain] and model (A) [price only] is reported per fold and in aggregate. A consistent positive delta across all three walk-forward folds constitutes evidence that on-chain signals are genuinely informative. An inconsistent or negative delta is reported as a null result.

Regression sub-task (30-day forward return):

- MAE and directional accuracy (sign agreement). Directional accuracy above 55% sustained across all three walk-forward folds constitutes a weak positive result; a result indistinguishable

from 50% confirms the efficient-market null hypothesis.

■ *All threshold choices and evaluation decisions are pre-registered in this proposal and will not be modified after results are observed. No performance threshold will be retroactively adjusted to make results appear favorable.*

7. Milestones and Deliverables

Week	Milestone	Concrete Deliverable	Risk Mitigation
1	Data pipeline & EDA	Merged daily CSV; EDA notebook: class distribution, ACF plots, feature correlation heatmap, missing-data audit	CoinMetrics bulk CSV as fallback if API rate-limited
2	Label engineering & feature construction	Labeled dataset under 3 threshold variants; rolling z-score normalization; lookahead-bias assertion suite	Sensitivity analysis built in — no single 'correct' threshold required
3	Baseline models (Ridge + Linear SVM)	Walk-forward F1 / MAE per fold; price-only vs. price+on-chain comparison table; coefficient analysis	Simplest models — highest confidence of completion
4	Nonlinear models (Random Forest, XGBoost)	Hyperparameter grid-search results; SHAP summary plots; feature importance ranking across folds	sklearn APIs well-documented; fallback to RF only if XGBoost install fails
5	Temporal CNN (PyTorch)	Training loss / val-F1 curves; test F1 per fold; full model comparison table across all four approaches	Scope reduced to MLP if CNN exceeds time budget — still satisfies neural network requirement
6	Report & code cleanup	Final PDF report; reproducible zip with requirements.txt and Colab-runnable notebooks; README with step-by-step instructions	Writing begins in Week 4 alongside modeling to avoid deadline compression

Scope is calibrated for a single graduate student over six weeks. Every milestone includes an explicit fallback. The write-up begins in parallel with Week 4 modeling so that results sections are drafted incrementally, not under deadline pressure.

References

- Ardia, D., Bluteau, K., & Rüede, M. (2019). Regime changes in Bitcoin GARCH volatility dynamics. *Finance Research Letters*, 29, 266–271. <https://doi.org/10.1016/j.frl.2018.08.009>
- Ballings, M., Van den Poel, D., Hespeels, N., & Gryp, R. (2015). Evaluating multiple classifiers for stock price direction prediction. *Expert Systems with Applications*, 42(20), 7046–7056. <https://doi.org/10.1016/j.eswa.2015.05.013>
- Rafi, S. et al. (2025). Using machine and deep learning models, on-chain data, and technical analysis for predicting bitcoin price direction and magnitude. *Expert Systems with Applications*. <https://doi.org/10.1016/j.eswa.2025.130875>
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All references verified against published sources before submission. Code submitted as a Python package;
notebooks runnable end-to-end on Google Colab.